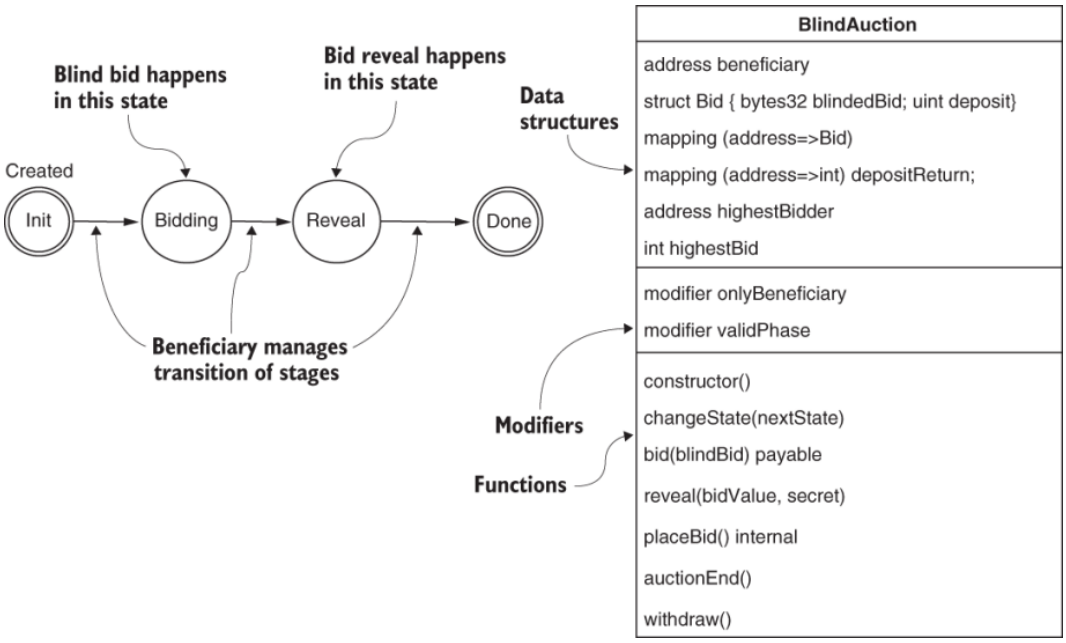



**Figure 5.11 State transition FSM and contract diagram for a blind auction**

## 5.4.2 Blind auction smart contract

The smart contract for the blind auction is a modified version of the code in the Solidity documentation, enabling us to focus on privacy and security concerning only one item: `blindedBid`. Let's review the code and then use the Remix IDE to test the smart contract. You can copy the listings to the Remix IDE as directed and explore it. Let's look at the following:

- Data elements (listing 5.3)
- Modifiers (listing 5.4)
- Functions (listing 5.5)

The data elements include data for the bid, an enumerated data type for the phases (states) of the auction, the beneficiary address, mappings for bids and deposit returns, and the highest bid details. These elements are defined in the next listing. Review the code, and load it into Remix.